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— *as to the applicant's entitlement to claim the priority of the earlier application (Rule 4.17(iii))*

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[Continued on next page]

(54) Title: SMART WASTE COLLECTION SYSTEM AND METHOD

(57) Abstract: A waste collection system (10) includes one or more waste containers (20) for receiving waste (50), a server system (100) for receiving one or more signals via a wireless communication network (70) from sensor arrangements (30, 40) included on the one or more waste containers (20) for sensing a waste status of the one or more containers (20). Beneficially, the server system (100) is operable from receiving the one or more signals to compute an optimal strategy for one or more waste collection vehicles (150) to collect waste from the one or more waste containers (20). The waste collection system (10) is capable of supporting a competitive bidding arrangement for implementing collection of waste (50) for the one or more waste containers (20).

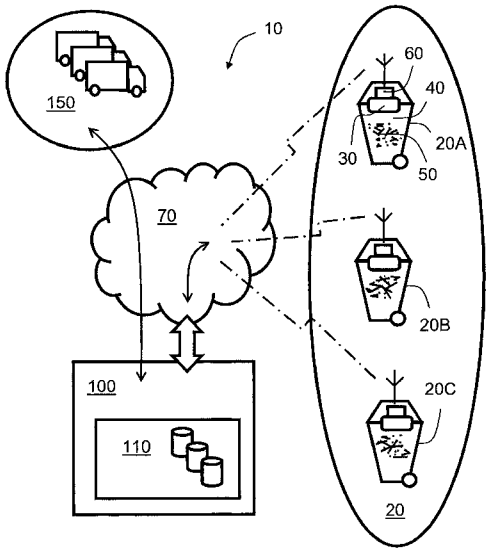


FIG. 1

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SMART WASTE COLLECTION SYSTEM AND METHOD

Technical field

The present disclosure relates to smart waste collection systems, for example to smart waste collection systems including one or more waste containers, also known as "trash containers", including a wireless-enabled monitoring arrangement, and a control centre for receiving waste-indicative signals from the one or more waste containers for devising an optimal schedule for one or more waste collection vehicles to collect waste from the one or more waste containers. Moreover, the present disclosure concerns methods of waste collection, for example to methods of monitoring one or more waste containers including a wireless-enabled monitoring arrangement, and operating a control centre for receiving waste-indicative signals from the one or more waste containers for devising an optimal schedule for one or more waste collection vehicles to collect waste from the one or more waste containers. Furthermore, the present disclosure relates to software products recorded on non-transient machine-readable data storage media, wherein the software products are executable upon computing hardware for implementing aforesaid methods.

Background

In nature, populations of animals grow in proportion to available food supply; when populations of animals grossly exceed their available food supply, population collapse often results. In the case of the human population, a similar growth trend has occurred, on account of enhancements in methods of food production and availability of sources of energy, for example fossil hydrocarbon fuels, for providing energy for farming and food distribution. The human population is presently around 7 billion people and growing at an approximately exponential trajectory as a function of time. An eventually human population collapse from circa 9 billion people to around 500 million to 1 billion people is expected to occur at a point in time in the future as effects of "peak oil" begin to impact economies of technologically-advanced societies, and energy-per-capita begins to reduce to non-sustainable levels; such a scenario is elucidated in a publication "*The Olduvai Theory: Energy, Population and Industrial Civilization*", Dr. Richard C. Duncan, Winter 2005-2006, J. Social Contract. "*Agenda*

21" (United Nations initiative) is concerned with managed human population reduction to sustainable numbers.

In technologically-advanced societies, for example as a result of mechanisation in farming, a relatively smaller portion of human population is required to execute functions of food production and food distribution, enabling a remainder of the human population to concentrate on other activities, often within urban environments. From United Nations statistics, soon over 50% of World human population will be living in urban environments.

Human activity creates waste, wherein such waste needs to be removed from urban environments in order to avoid a disruption of orderly functioning of such urban environments. As human population grows as aforementioned, existing resources become divided amongst ever more people, such that an increase in operating efficiency of human society is needed if a standard of living enjoyed by people is to be maintained in future. Operating efficiency of human society can be increased by employing recycling, wherein waste in itself becomes a potential resource. However, recycling activities themselves require resources, for example hydrocarbon fossil fuel for propelling waste collection vehicles, and salaries of waste collection staff which are subsequently used by to buy products and services requiring resources for their implementation. Thus, it is important, for a sustainable human population, that waste recycling activities are implemented in such a manner that they provide a net real benefit to the population.

The exponential growth in urban human population, the development of social economy, and improvements in human living standards have resulted in a significant increase in the amount of waste generation. It has thus been necessary to develop new technologies which aid efficient management of waste in urban environments. More recently, urban waste has been viewed as a resource, especially when its materials can be recycled, thereby avoiding environmental damage resulting from primary resource extraction activities; for example, urban waste includes many organic materials which can be bio-converted to peat-like materials, and many combustible materials which can be employed as a source of heating fuel in communal incinerators, for example in combined heat-and-power facilities.

In order that urban waste can be most beneficially recycled and/or disposed of, it is desirable that waste disposal methods are as efficient as possible in relation to resource utilization, for example energy utilization and personnel resource utilization.

In a published United States patent no. US 7, 957, 937B2 ("*Systems and methods for material management*"; Applicant – WM Trash Monitor Plus; Inventor – Waitkus), there is described a system and method for scheduling the emptying or replacement of a waste container based upon a degree to which the container is filled with waste, or a pattern of usage of the container. Such factors are considered to predict when the waste container will become completely full, and thus requiring to be emptied. Moreover, the system and method are operable to consider customer preferences and/or limitations of a waste hauler which is utilized to empty the waste container; the system and method determine, based upon the factors, an optimal time for the waste container to be emptied or replaced by the waste hauler. Furthermore, the factors are also used to determine when to accomplish suitable scheduling, namely when to notify the waste hauler that the waste container should be emptied or replaced at a given time. The method employs a computerized scheduling sub-system for scheduling purposes.

Smart waste containers are known; for example, in a published United States patent application no. US2009/0126473A1 ("*Method and device to indicate the content of garbage cans and vessels*"; Inventors – Porat, Havosha, Shvarzman and Katan), there is described a measuring arrangement for measuring the content of vessels and relay that information to persons remote from the vessels. The measurement arrangement is implemented, for example, as volume sensors, photo-detectors or lasers. The relay of information is optionally implemented via wires or wirelessly. Beneficially, the vessels are garbage cans. A display of display included on the garbage is used to indicate to users a volume of garbage in the garbage can. Optionally, wireless transmission of the volume information to a remote receiver is implemented, wherein the volume information is translated to a readable format so that garbage collectors are able to receive an overview of which garbage cans to empty and which may be left until a following collection.

Although aforesaid systems and apparatus for smart waste container collection are known, there exists a need for a waste collection system which is more optimized for the collection of waste in urban environments.

Waste management industries are growing and need efficient processes to increase revenue margins and to optimize associated resource utilization. From data provided in "*Environmental Business International*" publication, the US solid waste industry has grown from a value of 39.4 billion US dollars in the year 2000 to a value of 52.40 billion US dollars in the year 2010.

Waste collection companies face various challenges when implementing collection of waste from various sites and recycling stations at different locations; the challenges include the following, for example:

(i) planning and scheduling routes for waste haulers to employ for ensuring maximum waste collection;

(ii) avoiding penalties, for example fines from municipal authorities, for delayed collection of waste, for example where waste overflows from waste containers and potentially represents a safety and/or health hazard;

(iii) predicting customer waste generation patterns, for example based upon daily usage of waste containers, or during festivals and weekends when increased customer consumption of resources, for example food and drinks products, occurs; and

(iv) saving resources, and hence money, and reducing environmental impact of waste collection processes, for example less fuel consumption in waste collection vehicles, using less waste collection equipment, and optimizing waste collection intervals.

For example, it is highly inefficient for a waste collection vehicle to travel to a given site to empty a waste container which is only partially full of waste. It is desirable to improve an efficiency of waste collection, so that fewer resources are utilized in waste collection.

Summary

The present invention seeks to provide an improved smart waste collection system.

Moreover, the present invention seeks to provide an improved method of smart waste collection.

According to a first aspect, there is provided a waste collection system as defined in appended claim 1: there is provided a waste collection system including one or more waste containers for receiving waste, a server system for receiving one or more signals via a wireless communication network from sensor arrangements included on the one or more waste containers for sensing a waste status of the one or more containers, characterized in that the server system is operable from receiving the one or more signals to compute an optimal strategy for one or more waste collection vehicles to collect waste from the one or more waste containers.

The invention is of advantage in that collection of waste can be optimized.

The invention enables waste producers and collection operators to optimize waste collection quantities, waste collection routes adopted by waste collection vehicles and enhanced utilization of waste collection vehicles by, for example, only collecting full waste containers and thereby reducing unnecessary effort and resource utilization when collecting waste, optionally also recycling waste.

Moreover, the invention is capable of saving money for waste collection services, and is also capable of reducing an environmental impact of waste collection and recycling, for example by using less fuel and equipment for executing waste collection.

Furthermore, the invention is capable of improving safety and hygiene as full waste containers do not go unnoticed when the waste collection system is employed, namely no waste container overflow problems occur.

Additionally, the present invention is capable of providing live status information about monitored waste containers served by the waste collection system, for

example fill-up rates of waste containers, estimated fill-up dates of waste containers, scheduled waste collection dates for waste containers, and seasonal variations which may affect waste containers.

Additionally, the present invention is capable of encouraging recycling of waste based on measured changes in recycling behaviour.

Optionally, in the waste collection system, the one or more sensor arrangements of the one or more containers are arranged for determining whether or not the waste is in a fermenting and/or biodegraded state which could represent a health hazard, and for determining and sending corresponding information to a server system controlling the waste collection system, and the one or more sensor arrangements include a combination of a temperature sensor and a methane sensor to measure a level of fermentation and/or biodegradation of the waste.

Optionally, in the waste collection system, the server system is operable to employ Monte Carlo simulation to search a multi-dimension search space defined by parameters (A) influencing the one or more waste containers to determine the optimal strategy. More optionally, in the waste collection system, the one or more waste collection vehicles are provided with vehicle sensor arrangements for sensing spatial locations of the one or more vehicles and their waste capacity status in real-time for dynamically modifying the optimal strategy in real-time during collection of the waste from the one or more waste containers.

Optionally, in the waste collection system, the sensor arrangements of the one or more containers each include a waste status sensor arrangement for determining a volume of waste in a corresponding waste container, and a spatial position determining sensor for determining a spatial location of the corresponding waste container. More optionally, in the waste collection system, the sensor arrangement includes a user-actuated input sensor for enabling a user to signal to the server system that collection of waste of the container associated with the sensor arrangement is required to have waste collected therefrom.

Optionally, in the waste collection system, the server system includes a user portal for enabling competitive bidding from operators of the one or more vehicles to collect waste from one or more of the waste containers.

Optionally, in the waste collection system, the server system is arranged:
to generate one or more signals to compute a plurality of job offers for receiving a plurality of bids from the one or more operators of the one or more vehicles to collect waste from one or more of the waste containers;

to receive one or more signals from the one or more operators of the one or more vehicles for sensing one or more vehicle status of the one or more vehicles;

to select a lowest bid from the plurality of bids provided by the one or more operators of the one or more vehicles to collect waste from one or more of the waste containers;

to notify the one or more operators of the one or more vehicles of the selection of the lowest bid; and

to track the job offer until collection of waste from the one or more of the waste containers.

More optionally, in the waste collection system, one or more signals are used to compute a plurality of job offers, wherein the plurality of job offers is based on signals provided from the waste status sensor arrangement and the spatial position determining sensor of container and vehicle type and status.

More optionally, in the waste collection system, the system is arranged to receive vehicle status of the one or more operators of the one or more vehicles, wherein the vehicle status is pre-entered by the one or more operators of the one or more vehicles.

According to a second aspect, there is provided a waste container for use with the waste collection system pursuant to the first aspect: there is provided a waste

container including a sensor arrangement for determining a waste status of the waste container and sending corresponding information to a server system controlling the waste collection system.

According to a third aspect, there is provided a method of operating a waste collection system including one or more waste containers for receiving waste, a server system for receiving one or more signals via a wireless communication network from sensor arrangements included on the one or more waste containers for sensing a waste status of the one or more containers, characterized in that the method includes:

(a) operating the server system to receive the one or more signals from the sensor arrangement; and

(b) computing from the one or more signals an optimal strategy for one or more waste collection vehicles to collect waste from the one or more waste containers.

Optionally, the method includes operating the server system to employ Monte Carlo simulation to search a multi-dimension search space defined by parameters (A) influencing the one or more waste containers to determine the optimal strategy. More optionally, the method includes providing the one or more waste collection vehicles with vehicle sensor arrangements for sensing spatial locations of the one or more vehicles and their waste capacity status in real-time for dynamically modifying the optimal strategy in real-time during collection of the waste from the one or more waste containers.

Optionally, the method includes:

(d) operating the one or more sensor arrangements of the one or more containers to determine whether or not the waste is in a fermenting and/or biodegraded state which could represent a health hazard;

(e) determining and sending corresponding information to the server system controlling the waste collection system; and

(f) arranging for the one or more sensor arrangements to include a combination of a temperature sensor and a methane sensor to measure a level of fermentation and/or biodegradation of the waste.

Optionally, the method includes incorporating in each of the sensor arrangements of the one or more containers a waste status sensor arrangement for determining a volume of waste in a corresponding waste container, and a spatial position determining sensor for determining a spatial location of the corresponding waste container. More optionally, the method includes arranging for the sensor arrangement to include a user-actuated input sensor for enabling a user to signal to the server system that collection of waste of the container associated with the sensor arrangement is required to have waste collected therefrom.

Optionally, the method includes providing via the server system a user portal for enabling competitive bidding from operators of the one or more vehicles to collect waste from one or more of the waste containers.

According to a fourth aspect, there is provided a software product recorded on machine-readable data storage media, characterized in that the software product is executable upon computing hardware for implementing a method pursuant to the third aspect.

It will be appreciated that features of the invention are susceptible to being combined in various combinations without departing from the scope of the invention as defined by the appended claims.

Description of the diagrams

Embodiments of the present disclosure will now be described, by way of example only, with reference to the following diagrams wherein:

FIG. 1 is an illustration of a waste collection system implemented pursuant to the present disclosure;

FIG. 2 is an illustration of an example route followed by a waste collection vehicle to collect waste from a plurality of waste containers disposed at various spatial locations within an urban environment; and

FIG. 3 is a mapping employed when performing optimization computations for the waste collection system of FIG. 1; and

FIG. 4 is an illustration of steps of a method for a bidding process of the disclosure.

In the accompanying diagrams, an underlined number is employed to represent an item over which the underlined number is positioned or an item to which the underlined number is adjacent. A non-underlined number relates to an item identified by a line linking the non-underlined number to the item. When a number is non-underlined and accompanied by an associated arrow, the non-underlined number is used to identify a general item at which the arrow is pointing.

Description of embodiments of the invention

Referring to FIG. 1, there is shown an illustration of a waste collection system indicated generally by **10**. The system **10** includes one or more waste containers **20**, namely "trash cans". Each waste container **20** includes, for example in its pivotally-mounted lid, a sensor arrangement **30** for monitoring one or more physical parameters of an inside volume **40** within the waste container **20**, wherein the inside volume **40** is designed to receive waste. The sensor arrangement **30** includes one or more sensors for monitoring an extent to which the inside volume **40** is occupied by waste **50**. Optionally, the sensor arrangement **30** includes a flammable gas sensor for measuring whether or not the waste **50** is in a fermenting and/or biodegrading state which could represent a health hazard; beneficially, the gas sensor is implemented according to at least one of:

(i) a solid-state methane sensor;

(ii) an electro-chemical gas concentration sensor;

(ii) a pulsed hot-wire pellistor sensor; and

(iii) an infra-red absorption sensor, for example based upon a MEMs tunable radiation filter for enable one of more infra-red active gas concentration to be measured; optionally active electronic parts of a sensor are disposed outside the container, and an optical gas-absorption part of the sensor is implemented within the container **20**.

Optionally, the sensor arrangement **30** includes a position sensor, for example a position sensor based upon GPS or GPRS measurement, for determining a spatial location of the waste container **20** within an urban environment. The waste container **20** also includes a wireless interface **60** coupled to the sensor arrangement **30** for

receiving sensor signals therefrom; the wireless interface **60** is operable to communicate the sensor signals via a wireless communication network **70**, for example GSM or 3G implemented, to a server system **100**; the server system **100** is thus beneficially provided with periodic information, for example substantially real-time information, regarding an extent to which the one or more waste containers **20** are filled with waste, optionally also with an indication of a nature of the waste, and optionally spatial position information pertaining to the one or more waste containers **20** within the urban environment. Optionally, each waste container **20** is provided with a unique identification code by which the waste container **20** can be identified, wherein the unique identification code is also communicated via the wireless interface **60** to the server system **100**. Information communicated from the one or more waste containers **20** to the server system **100** is stored in a database **110** associated with the server system **100**.

The server system **100** is operable to maintain a simulated model of the one or more waste containers **20**, namely their spatial locations within the urban environment, their degree of waste **50** filling their inside volume **40**, a nature of waste in the waste containers **20** and a temporal rate at which they are being filled with waste **50**. Moreover, the database **110** includes information regarding temporal patterns which modulate rates at which are generated by customers associated with the one or more waste containers **20**; the temporal patterns optionally take into account holidays, festivals, and weather conditions for example; for example, biological waste which is susceptible to rapid oxidation at high ambient temperatures beneficially requires more frequent collection in order to avoid potential health hazards, for example spreading of dangerous microbes; for example, the sensor arrangement **30** also includes a temperature sensor for enabling the server system **100** to receive data indicative of temperatures within the one or more waste containers **20**.

Moreover, biological waste potentially at times starts to break down, for example by way of fermentation of food and other bio-waste, such as sludge, which results in a break down of biodegradable material, for example as a result of alternating or changing conditions within the container **20**. These changing conditions in the container **20** are measured and monitored over time by monitoring one or more of:

(i) a temperature within the container **20**;

(ii) any gas generation occurring within the container **20**;

(iii) a pressure within the container **20**;

(iv) a colour of contents within the container **20**;

(v) a density of contents within the container **20**;

but not limited thereto. Break down processes occurring within the container **20** generates gases such as Hydrogen, methane and other hydrocarbon gases that potentially cause several unwanted problems in the environment, as well as health concerns, unwanted smells, explosion risks, and attracting insects and animals dependent upon where the container **20** is placed, By monitoring and measuring the content of the container **20**, any generated gases or changing conditions beneficially allow for changes in a method employed when, for example, collecting the waste material, for example for use in biogas production through an anaerobic digestion process. Such biogas production also results in bad odor and air quality around the container **20**, for example around restaurant premises, kitchens, yards. Moreover, such biogas production potentially results in structural deformation of the container **20** due to gases expanding, where there is a lack of sufficient ventilation from the container **20**. Furthermore, such biogas production results in fire hazards and also affects a quality of the material within the container **20**, for example in a situation where the material has started to ferment heavily, wherein the material cannot be used as efficiently in an anaerobic digestion plant for biogas production.

Beneficially, the waste collection system **10** also includes a sensor arrangement **30**, **40** for determining a waste status of the waste container **20**. Optionally, the sensor arrangement **30**, **40** sends corresponding information to a server system **100** controlling the waste collection system **10**, wherein the sensor arrangement **30**, **40** is operable to monitor at least one of:

(i) methane concentration;

(ii) Carbon Dioxide concentration;

(iii) Carbon Monoxide concentration,

and other hydrocarbon gases generated during fermentation. Optionally, the sensor arrangement **30**, **40** consumes negligible power and is operable over a wide range of temperature variations, in high humidity and in condensation areas. Optionally, a part of the sensor for measuring biodegradability of the waste **50**, such as a gas sensor, is optionally located as a modular construction or addition to the sensor arrangement

30, **40**, thereby allowing it to be added to a waste collection system with one or more sensors per container **20** when and if needed. Optionally the gas sensor could be arranged as an integrated part of the sensor arrangement **30**, **40**, as a complementing sensor next to the sensor arrangement **30**, **40**, or in a different position than the sensor arrangement **30**, **40** if more appropriate to measure small changes in an environment within the container **20**, or biodegradability of the waste **50** within the container **20**. In another embodiment, the gas sensor id disposed outside the waste container **20** to monitor more than one container **20** at a time, namely the gas sensor is common to a plurality of the containers **20**. The gas sensor, namely biodegradability sensor, is beneficially also used to measure the level of biodegradable material in the container **20**, wherein, in most cases, the container **20** beneficially has a safety valve arranged in the upper part of the container **20** to cope in the event of dangerous levels of gas generation within the container **20**, A commercial value of the container **20**, and its waste **50**, namely content, for pick up can be calculated based on a nature of the content, a level of fill of waste 50 within the container **20**, a size of container **20**, and a level of biodegradability of the content, namely waste **50**.

The server system **100** is operable to compute an optimal strategy for a fleet of vehicles **150**, for example one or more waste collection vehicles, to service the one or more waste containers **20**, for example for maximizing profit for a refuse collection service or for reducing operating costs for the refuse collection service, whilst complying with safety requirements and service quality criteria to avoid imposition of fines and penalties for allowing one or more of the waste containers **20** to become overfilled and/or their waste **50** to reach a physical state that could represent a safety hazard (for example spread of disease and pests (for example rodents) arising from food wastes).

The optimal strategy devised by the server system **100**, wherein the server system **100** includes computing hardware which is operable to execute one or more software products stored on machine-readable data storage media, is communicated to the fleet of vehicles **150** for guiding their route of travel for reaching the one or more containers **20**, for removing their waste **50** into the vehicles **150** and transporting the waste **50** to one or more waste disposal or waste recycling facilities **160**. Optionally,

the optimal strategy includes devising a most efficient route **170** for a given vehicle **150** to take when servicing a plurality of waste containers **20** within the urban environment as depicted in FIG. 2.

Computing the optimal strategy is a non-trivial task because of a multi-dimensional nature of a solution space for devising the optimal strategy. Factors which can influence efficiency of performance of the fleet of vehicles **150** include one or more factors as given in Table 1:

Table 1: influencing factors

Influencing factor	Factor detail
A_1	a spatial location of a given waste container **20**, for example expressed as a distance from a temporally previous container **20** along a route to the given waste container **20**
A_2	a nature and state of the waste **50** included in the given waste container **20**
A_3	a rate at which the given waste container **20** is filling with waste **50**
A_4	a date on which the waste **50** in the given waste container **20** is to be collected, for example in respect of day in calendar year, holidays, festivals and so forth
A_5	a degree of access of a given vehicle **150** to the given waste container **20**, for example poor access for the vehicle **150** to reach the given container **20** increases a time required for the vehicle **150** to service to the given container **20**
A_6	a potential penalty which may pertain to the given container **20**, depending upon a nature of the waste **50** in the given container **20**
A_7	traffic conditions which pertain along a route of travel of the vehicle **150** when travelling to reach the given waste container **20**, for example traffic jams, scheduled road repairs, vehicle breakdowns, and so forth

The factors in Table 1 are not exhaustive, such that the waste collection system **10** can take into account fewer or more factors when performing its optimization computations as will be described in more detail later.

The server system **10** is operable to maintain a model of the urban environment in with the waste collection system **10** is required to provide waste collection services, as well as well as representations C(*i*) of waste containers **20**, wherein *i* is an index which uniquely defines each waste container **20**, wherein $i = 1, 2, 3, \ldots n$, wherein *n* is a total number of waste containers **20** to be serviced in the urban environment.

The server system **100** is operable to implement a Monte Carlo analysis of a cost function F and a waste value function G for all combinations of vehicle **150** travel between the waste containers **20** that are to be serviced. This requires mapping the containers C(*i*) for *i* = 1 to *n* to a test array K(*j*) as depicted in FIG. 3 via an order mapping function **180**, corresponding to different possible routes of travel for the fleet of vehicles **150**. The cost function F and the waste value function G, namely a financial value of the waste when recycling is pertinent, are pursuant to Equation 1 (Eq. 1) and Equation 2 (Eq. 2):

$$F = \sum_{j=1}^{m} \left(f\left(K_j \left(A_{j,1}, A_{j,2}, A_{j,3}, \ldots A_{j,q} \right) \right) \right) \qquad \text{Eq. 1}$$

wherein

m = an index in a range of 1 to *n*;

q = factor type reference index;

f = collection cost function; and

F = total cost of making a collection of containers K in a range 1 to *m*.

Similarly, a recycling value of the waste **50** collected by the fleet of vehicles **150** can be computed from Equation 2 (Eq. 2):

$$G = \sum_{j=1}^{m} \left(g\left(K_j \left(A_{j,1}, A_{j,2}, A_{j,3}, \ldots A_{j,q} \right) \right) \right) \qquad \text{Eq. 2}$$

wherein

m = an index in a range of 1 to *n*;

q = factor type reference index;

g = waste value function; and

G = total value from making a collection of containers K in a range 1 to *m*.

The order mapping function **180** enables different routes between the waste containers **20** to be simulated by the computation in the server system **100**; all combination of routes are beneficially explored, even if some would be regarded as potentially inefficient and subsequently ignored for optimization purposes to save computing effort in the server system **100**.

The server system **100** is operable to compute optimal values of the functions F and G as a function of the index *m* and order mapping function **180**. When an optimal value for the functions F and G are found, the route as defined by the order mapping function **180** is then translated to a form suitable for instructing drivers of the fleet of vehicles **150** for executing a collection of waste **50** from selected one or more of the waste containers **20**.

Optionally, during an execution of collection of the waste **50**, the one or more vehicles **150** are equipped with GPS, or similar position detection sensors, and wireless interfaces for providing in real time an indication of positions of the one or more vehicles **150** to the server system **100**, together with an indication of remaining waste holding capacity of the one or more vehicles **150**, such that the server system **100** is operable to perform real time Monte Carlo computations, as aforementioned, to make optimization corrections to routes of travel of the one or more vehicles **150**, for example in an event of one or more of the vehicles **150** being delayed in their collection round, one or more of the vehicles **150** having a breakdown and needing repair, one or more of the vehicles **150** having unexpectedly more waste collection capacity remaining, and so forth, so that the collection of waste **50** can be optimized in real time as environmental conditions influence the collection of the waste **50** from one or more of the containers **20**. Optionally, vehicles **150** of the fleet are specialized in handling certain types of waste **50**, and their routes are computed as aforementioned on a basis of containers **20** which are of a type that they are permitted to service for waste collection purposes.

The functions G, g are dependent upon a price for waste **50,** namely price for "trash", for example metal and paper waste, which can be sold for reuse. Moreover, the functions F, f are dependent upon the cost of disposing of various types of, for example, hazardous toxic waste. The computation model executed in the server system **100** also computes an expected amount of waste **50** in the containers **20** as a function of time, taking into account waste collection activities in respect of the containers **20**, to ensure that they do not become overfilled, thereby seeking to avoid imposition of penalties for over-filled containers **20**. Optionally, the containers **20** are provided with a user-operable "please empty me" switch which users can actuate to

send a message wirelessly to the server system **100** that emptying of their containers **20** is desirable, for example in anticipation by the users of activities being undertaken by the users that are likely to generate considerable waste, for example building renovation activities, pop-concerts, rave parties and so forth.

Beneficially, the server system **100** is operable to support a bidding system provided via, for example, at least one of: Internet, cellular communication networks, mobile applications and so forth; beneficially, a plurality of waste collection enterprises or individuals, each with its associated fleet of one or more vehicles **150**, is able to bid competitively for executing collection of waste **50** from one or more of the containers **20**, pursuant to an optimized collection plan proposed by the server system **100** as determined using Table 1, and Equations 1 and 2.

Optionally, the server system **100** collects a plurality of information from the sensors attached to the containers **20**. The plurality of information relates to information such as accurate prediction that the containers are likely to be full, for example filled to 90% of their capacity, container type, container quantity at each location, waste fraction type, accurate prediction on the volume and weight of all waste in the containers, value of the waste (price per tonne), location of each container, distance between each location, and so forth. The plurality of information collected from the containers **20** is processed to generate a plurality of job offers for the plurality of waste collection enterprises. Depending upon a type of vehicle available from the waste collecting enterprise, the operator of the server system **100** beneficially implements a reverse auction, wherein those vehicles within range of an optimised collection route of well filled containers **20** of desirable type and quantity of waste for given vehicles are requested whether or not they want to execute a waste collection job and for what price. Optionally, the job offer displays information such as optimized route distance, waste value, total time required to complete the job, type of truck required, container type and waste fraction type and so forth. The plurality of job offer, for example, appears in a manner similar to that illustrated in Table 2.

Table 2: Example of Auction for executing a collection round

JOB OFFER ID: 231
Current bid: **122 USD**
Total distance: 32km
Material value: 227.7 USD
(30 USD / 1000kg)
Quantity: 23 m3, 7590kg
Total time (estimate): 230 minutes
(3h 40 minutes), 1/2 work shift
Waste fraction: Clear glass
Container type(s): 4, 6 and 8yd
front loading dumpsters

Date/Time: 12/11/2013, 12:09am
Time to bid left: 5 hours and 40 minutes

Optionally, the server system **100** provides a portal wherein the various enterprises are able to enter parameters relating to their fleet of vehicles **150**. The parameters pertain to details such as truck type (for example front loader, rear loader, side loader, crane), truck capacity, suitable waste fractions, what type of time slots the hauler can accept (2 hour, 4 hour or 8 hour jobs), geographical information about serviceable areas (geo fence), availability (time/date) and so forth. Beneficially, the portal allows various enterprises to bid on the plurality of job offers based on the information displayed by the plurality of job offers and the parameters relating to fleet of vehicles **150**, wherein the portal allows various enterprises to bid based on the location of a vehicle within the fleet of vehicles, preferably within a range of for example ca 200 km. In applications of job offers in Towns and Cities the range may be more in the range of a radius of up to ca 50km, or more preferably radius of up to ca 10km from the based depot. Optionally, the portal provides real time information on the job offer and generates cost/profit potential for the various enterprises based on the parameters related to the fleet of vehicles. An embodiment of the portal includes a dash board with suitable container **20** pickups and Return-on-Investment (ROI) on any displayed task with information as outlined in Appendix A below.

Optionally, the server system **100** selects the lowest bid for a job offer and notifies the enterprises about the selection and the vehicle from the fleet of vehicles **150** for executing collection of waste **50** from one or more of the containers **20.** The server system **100** beneficially tracks the plurality of job offers until completion of the job, wherein the tracking relates to providing route information to the vehicle, monitoring timing, monitoring an emptying event of each container **20**, thereby ensuring that the

job is complete. This allows competitive bidding from enterprises, such that the waste containers **20** are serviced for waste collection purposes in a most commercially efficient manner. The portal beneficially enables the waste-filled status of one or more of the waste containers **20** to be viewed by third parties, for example refuse collection enterprises, for example as a function of their unique identities, but not via their spatial location in order to maintain customer confidentiality.

Beneficially, the server system **100** also benefits the customer, wherein the customer pays for collecting waste from the waste containers **20.** The customer only pays for their share of the winning bid. The server system **100** considers one or more following factors for determining the share of the customer, for example:

(i) containers **20** that are over 80% full;

(ii) material value goes to the hauler;

(iii) cost of overhead driving (to/from depot/dump);

(iv) the exact driving cost/share for collecting customer specific containers.

An embodiment of the benefits accrued to the customer are, for example, found in Appendix A below.

Optionally, the server system **10** is operable to invoice customers associated with the waste containers **20** as a function, optionally also nature, of waste **50** collected from their respective waste containers **20**. Such invoicing is beneficially implemented automatically, for example via Internet communication such as e-mail, as a function of quantity of waste **50**, thereby potentially encouraging generation of less waste **50** and hence more efficient resource utilization in the urban environment served by the waste collection system **10**.

Contemporary known arrangements for collecting waste, for example agreed with municipal authorities, tend to be extremely static and employ fixed pricing arrangement, which stifles innovation and development of best practices in waste collection and waste recycling.

Optionally, the waste collection system **10** can be employed in conjunction with large undergrounds waste containers or surface containers, for example waste containers having an internal waste-receiving volume in a range of $1m^3$ to $10m^3$. The waste

collection system **10** is capable of providing increased hygiene, thereby reducing a risk of spread of disease within urban environments; from a viewpoint of biological considerations, urban environments represent a large monoculture of human beings which is potentially susceptible to systemic pathogens, for example bubonic plague which caused the Black Death in the mediaeval epoch in Europe, where 30% of human population was eliminated. More efficient collection of waste **50** from the one or more containers **20** by utilizing the present invention is thus capable of reducing a risk in future of occurrence of such disease outbreak by preventing vectors of disease, for example rodents, from increasing in numbers.

In an alternative embodiment the bidding process for containers may also be applied in reverse so that it applies to containers or storage means needing to be filled up with content such as for example grain, milk, water, oil, sand, ballast, or other material or liquid. Bids would then be put in to complete a job where a truck or lorry completes a delivery of product to a depot or storage facility. The monitoring of such storage means would therefore be set up to indicate when e.g. a container is empty or approaching to be empty. This monitoring of storage means that need to be refilled as they approach an empty level of for example only 25% full, 15% full or less than 10% full may be used in applications such as Manufacturing sites, Harbours, Airports, Agriculture, Mining, Refineries and similar.

Modifications to embodiments of the invention described in the foregoing are possible without departing from the scope of the invention as defined by the accompanying claims. Expressions such as "including", "comprising", "incorporating", "consisting of", "have", "is" used to describe and claim the present invention are intended to be construed in a non-exclusive manner, namely allowing for items, components or elements not explicitly described also to be present. Reference to the singular is also to be construed to relate to the plural. Numerals included within parentheses in the accompanying claims are intended to assist understanding of the claims and should not be construed in any way to limit subject matter claimed by these claims.

APPENDIX A:

An example of a collection bidding process for containers **20** to be collected in 3 days is herewith provided, referring to FIG. 4 which outlines the process of bidding for collection of one or more containers.

STEP 1

A customer 1 (C1) has 2 locations, a customer C4 has 3 locations and a customer C8 has 2 locations, so there are a total of 7 locations that need to have waste from their containers **20** be collected within 3 days.

The Enevo One Collect™ system is provided with following information for the aforesaid locations:

1. An accurate prediction that the containers are going to be full (for example at 90% of capacity);
2. Information regarding container type;
3. Information regarding container quantity at each location;
4. Information regarding waste fraction type;
5. An accurate prediction on a volume and weight of all waste in the one or more containers **20**;
6. A value of the waste (price / ton) shown as Vn in FIG. 4;
7. A spatial location of each container **20**; and
8. A distance between each location.

STEP 2

The Enevo One Collect™ system then generates a job offer based on the above 8 parameters and thereafter it posts the offer to an open pool wherein potential waste collection service providers are able to bid for winning the job. The job offer is open for bidding for a limited time only and includes, for example, following information:

1. An estimate on a total optimized route distance for servicing all the locations and picking up all full containers thereat;
2. An estimate on the waste value (namely a brokerage price), quantity (a volume and a weight);

3. An estimate on time required to do the job (for example calculated based on a driving time between the locations, and a container-type-based collection time, a preparation time at the locations and a route entry and a depot driving time);
4. A truck type required or suitable for the job;
5. A waste fraction type; and
6. Information regarding container type or types.

Example:

JOB OFFER ID: 231 Date/Time: 12/11/2013, 12:09am

Current bid: 122 USD Time to bid left: 5 hours and 40 minutes

Total distance: 32km

Material value: 227.7 USD (30 USD / 1000kg)

Quantity: 23 m3, 7590kg

Total time (estimate): 230 minutes (3h 40 minutes), 1/2 work shift

Waste fraction: Clear glass

Container type(s): 4, 6 and 8yd front loading dumpsters

STEP 3

Waste / Recycling collection companies (namely haulers) enter the Enevo One Collect™ system and are able to monitor all open job offers and their current bidding status. The haulers have pre-entered information regarding their collection fleet (namely trucks). The information entered is:

1. Truck type (for example front loader, rear loader, side loader, crane);
2. Truck capacity;
3. Suitable waste fractions;
4. What type of time slots the hauler can accept (2 hour, 4 hour or 8 hour jobs);
5. Geographical information about serviceable areas (geofence); and
6. Availability (time/date)

The haulers then can monitor all suitable job offers based on the pre-entered fleet information. If a suitable job offer is available, the hauler is able to enter a bid into the system. The hauler gets real information on the job and the Enevo system is then operable to calculate accurately their cost/profit potential based on given parameters.

STEP 4

The Enevo One Collect™ system handles the bidding process and selects the lowest bid for the job offer. The hauler winning the Engagement, shown as En in FIG. 4, takes on the job. The hauler is notified and the availability of the hauler's truck (namely suitable for this job) is locked for time/date required for the won job offer.

The Enevo One Collect™ system then calculates an optimal route for the hauler's truck to fulfil the job. The route information is calculated based on the pre-entered data on where the truck is located. This route optimization is a key parameter to take into account for implementing the job efficiently.

STEP 5

The Enevo One Collect™ system follows each job offer and keeps tracking that the job is completed. The tracking is implemented based on the route calculation provided to the hauler, monitoring the timing, monitoring the emptying event of each container and fill level after the container was collected. This gives assurance to the waste producing customer that the work has been implemented.

STEP 6

The winning bid fee is shared between the waste producing customers that had containers participating in the job offer. The fee is calculated and billed electronically (for example via credit card or PayPal) by the Enevo One Collect™ web system.

The fee is calculated based on a sum of following parameters:

1. A total overhead driving completed by the hauler on the job offer divided by the total sum of kilometres required to drive to collect the specific customers containers;
2. A total sum of kilometres required to drive to collect the customers' containers;
3. A time spent emptying the specific customers' containers (for example a 240 litre wheelie bin requires approximately 15 seconds for emptying, whereas a 8 cubic yard dumpster requires for 60 seconds for emptying);
4. A total quantity (namely volume and weight) picked up from the specific customers containers; and

5. Deducted with the value of the material.

STEP 7

The winning bid fee collected from the waste producing customers is then deposited to the waste collection company account. Before depositing the money to the hauler, the Enevo One Collect™ system deducts a commission/brokerage fee for each billed job offer. Other factors taken into consideration during the collection bidding include at least one of: labour costs, driving costs, a preparation time, an energy capacity of container content, a market index, a value of container content.

In an alternative embodiment the bidding process for containers may also be applied in reverse so that it applies to containers or storage means needing to be filled up with content such as for example grain, milk, water, oil, sand, ballast, or other material or liquid. Bids would then be put in to complete a job where a truck or lorry completes a delivery of product to a depot or storage facility. The monitoring of such storage means would therefore be set up to indicate when e.g. a container is empty or approaching to be empty.

CLAIMS

1. A waste collection system (10) including one or more waste containers (20) for receiving waste (50), a server system (100) for receiving one or more signals via a wireless communication network (70) from one or more sensor arrangements (30, 40) included on the one or more waste containers (20) for sensing a waste status of the one or more containers (20), characterized in that

the server system (100) is operable from receiving the one or more signals to compute an optimal strategy for one or more waste collection vehicles (150) to collect waste from the one or more waste containers (20).

2. A waste collection system (10) as claimed in claim 1, characterized in that the server system (100) is operable to employ Monte Carlo simulation to search a multi-dimension search space defined by parameters (A) influencing the one or more waste containers (20) to determine the optimal strategy.

3. A waste collection system (10) as claimed in any one of claims 1 and 2, characterized in that the one or more sensor arrangements (30, 40) of the one or more containers (20) are arranged for determining whether or not the waste (50) is in a fermenting state which could represent a health hazard, and for determining and sending corresponding information to a server system (100) controlling the waste collection system (10), and the one or more sensor arrangements include a combination of a temperature sensor and a methane sensor to measure a level of fermentation and/or biodegradation of the waste (50).

4. A waste collection system (10) as claimed in any one of claims 2 and 3, characterized in that the one or more waste collection vehicles (150) are provided with vehicle sensor arrangements for sensing spatial locations of the one or more vehicles (150) and their waste capacity status in substantially real-time for dynamically modifying the optimal strategy in substantially real-time during collection of the waste (50) from the one or more waste containers (20).

5. A waste collection system (10) as claimed in claim 1, 2, 3 or 4, characterized in that the sensor arrangements (30, 40) of the one or more containers (20) each include a waste status sensor arrangement for determining a volume of waste in a corresponding waste container (20), and a spatial position determining sensor for determining a spatial location of the corresponding waste container (20).

6. A waste collection system (10) as claimed in claim 5, characterized in that the sensor arrangement (30, 40) includes a user-actuated input sensor for enabling a user to signal to the server system (100) that collection of waste of the container (20) associated with the sensor arrangement (30, 40) is required to have waste (50) collected therefrom.

7. A waste collection system (10) as claimed in any one of the preceding claims, characterized in that the server system (100) includes a user portal for enabling competitive bidding from one or more operators of the one or more vehicles (150) to collect waste (50) from one or more of the waste containers (20).

8. A waste collection system (10) as claimed in claim 7, characterized in that the server system (100) is arranged:

to generate one or more signals to compute a plurality of job offers for receiving a plurality of bids from the one or more operators of the one or more vehicles (150) to collect waste (50) from one or more of the waste containers (20);

to receive one or more signals from the one or more operators of the one or more vehicles (150) for sensing one or more vehicle status of the one or more vehicles (150);

to select a lowest bid from the plurality of bids provided by the one or more operators of the one or more vehicles (150) to collect waste (50) from one or more of the waste containers (20);

to notify the one or more operators of the one or more vehicles (150) of the selection of the lowest bid; and

to track the job offer until collection of waste (50) from the one or more of the waste containers (20).

9. A waste collection system (10) as claimed in claim 8, characterized in that one or more signals are used to compute a plurality of job offers, wherein the plurality of job offers is based on signals provided from the waste status sensor arrangement (30, 40) and the spatial position determining sensor of container and vehicle type and status.

10. A waste collection system (10) as claimed in claim 8 or 9, characterized in that the system (100) is arranged to receive vehicle status of the one or more operators of the one or more vehicles (150), wherein the vehicle status is pre-entered by the one or more operators of the one or more vehicles (150).

11. A waste container (20) for use with the waste collection system (10) of claim 1, characterized in that the waste container (20) includes a sensor arrangement (30, 40) for determining a waste status of the waste container (20) and sending corresponding information to a server system (100) controlling the waste collection system (10).

12. A method of operating a waste collection system (10) including one or more waste containers (20) for receiving waste (50), a server system (100) for receiving one or more signals via a wireless communication network (70) from sensor arrangements (30, 40) included on the one or more waste containers (20) for sensing a waste status of the one or more containers (20), characterized in that the method includes:

(a) operating the server system (100) to receive the one or more signals from the sensor arrangement (30, 40); and

(b) computing from the one or more signals an optimal strategy for one or more waste collection vehicles (150) to collect waste from the one or more waste containers (20).

13. A method as claimed in claim 12, characterized in that the method includes:

(c) operating the server system (100) to employ Monte Carlo simulation to search a multi-dimension search space defined by parameters (A) influencing the one or more waste containers (20) to determine the optimal strategy.

14. A method as claimed in claim 12 or 13, characterized in that the method includes:

(d) operating the one or more sensor arrangements (30, 40) of the one or more containers (20) to determine whether or not the waste (50) is in a fermenting and/or biodegraded state which could represent a health hazard;

(e) determining and sending corresponding information to the server system (100) controlling the waste collection system (10); and

(f) arranging for the one or more sensor arrangements to include a combination of a temperature sensor and a methane sensor to measure a level of fermentation and/or biodegradation of the waste (50).

15. A method as claimed in claims 13 or 14, characterized in that the method includes providing the one or more waste collection vehicles (150) with vehicle sensor arrangements for sensing spatial locations of the one or more vehicles (150) and their waste capacity status in substantially real-time for dynamically modifying the optimal strategy in substantially real-time during collection of the waste (50) from the one or more waste containers (20).

16. A method as claimed in as claimed in any one of claims 12 to 15, characterized in that the method includes incorporating in each of the sensor arrangements (30, 40) of the one or more containers (20) a waste status sensor arrangement for determining a volume of waste in a corresponding waste container (20), and a spatial position determining sensor for determining a spatial location of the corresponding waste container (20).

17. A method as claimed in as claimed in any one of claims 12 to 16, characterized in that the method includes arranging for the sensor arrangement (30, 40) to include a user-actuated input sensor for enabling a user to signal to the server system (100) that collection of waste of the container (20) associated with the sensor arrangement (30, 40) is required to be have waste (50) collected therefrom.

18. A method as claimed in any one of claims 12 to 17, characterized in that the method includes providing via the server system (100) a user portal for enabling competitive bidding from operators of the one or more vehicles (150) to collect waste (50) from one or more of the waste containers (20).

19. A software product recorded on non-transient machine-readable data storage media, characterized in that the software product is executable upon computing hardware (100, 110) for implementing a method as claimed in any one of claims 12 to 18.

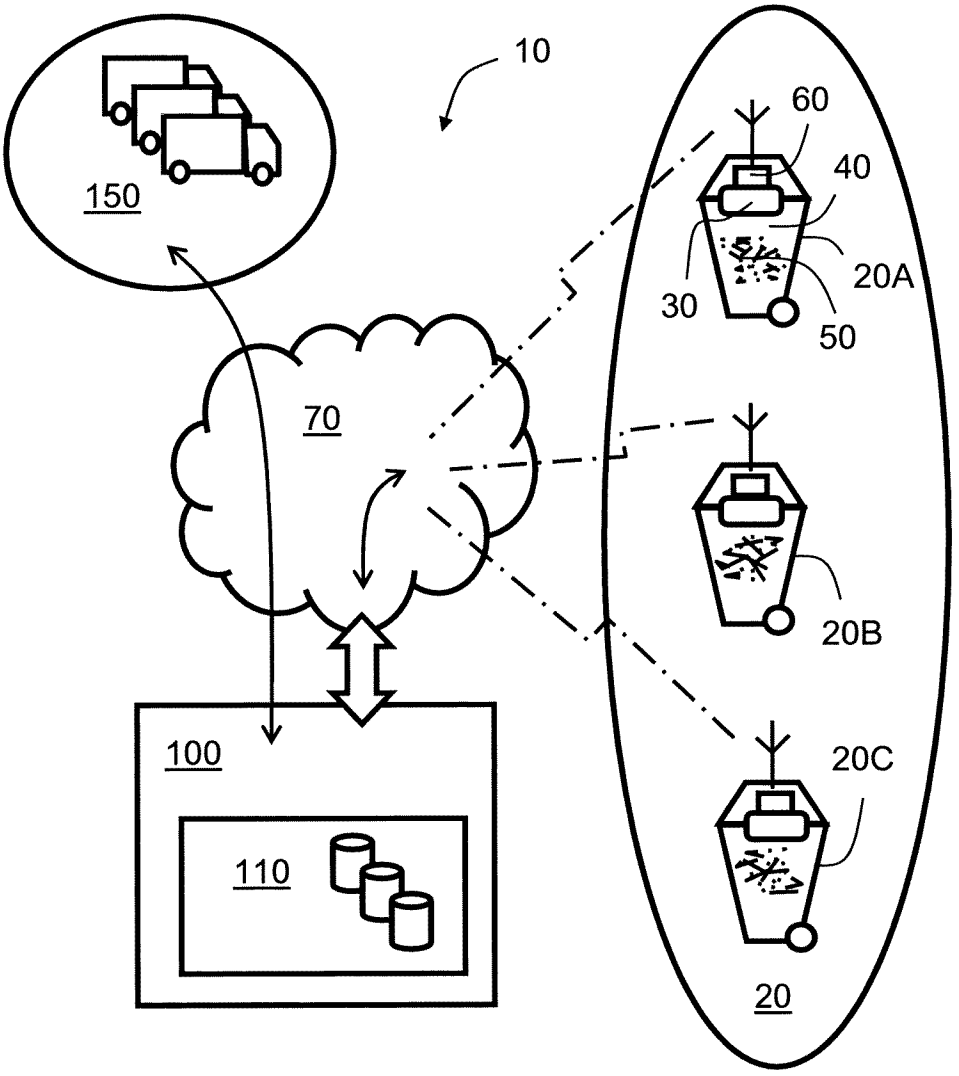


FIG. 1

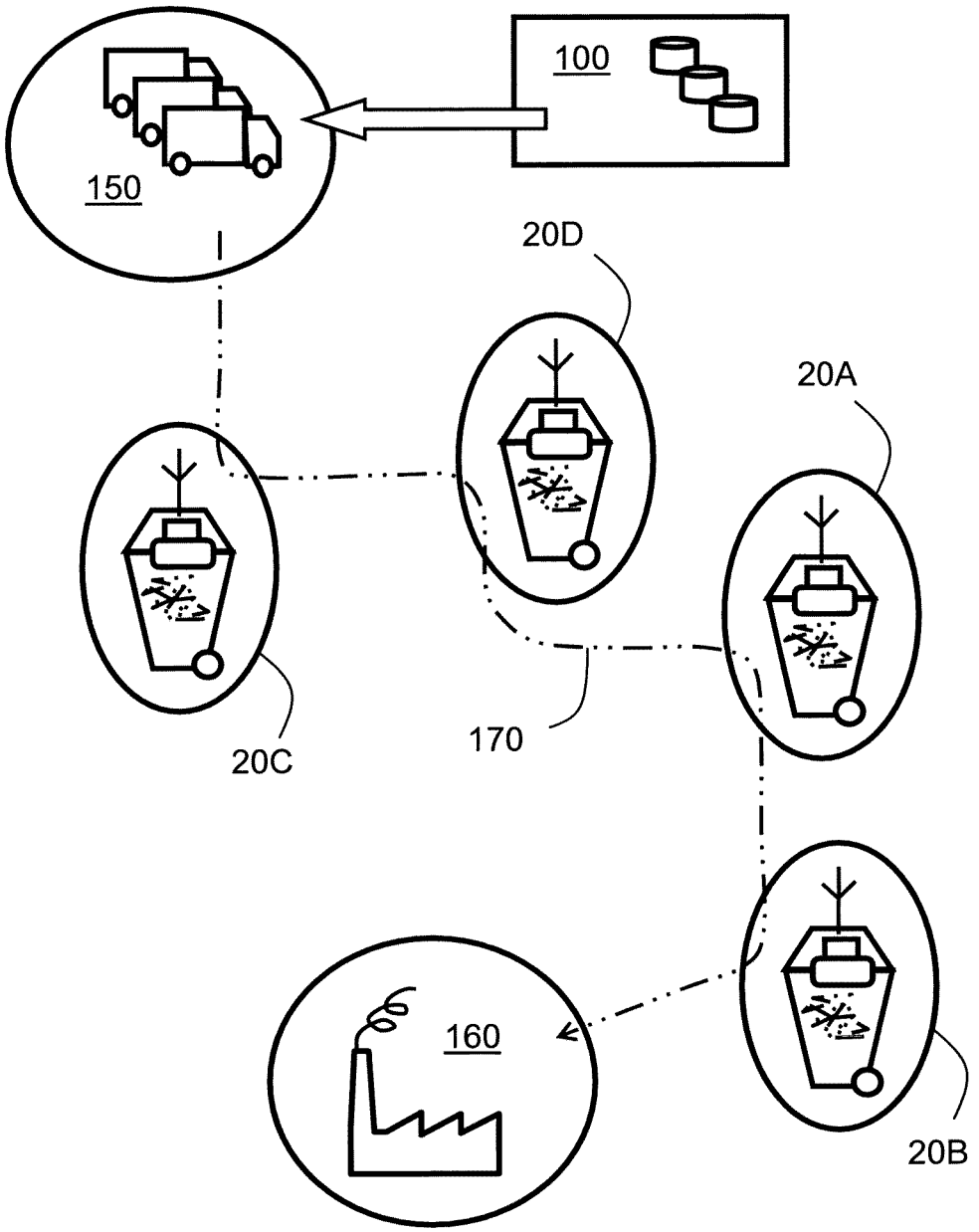


FIG. 2

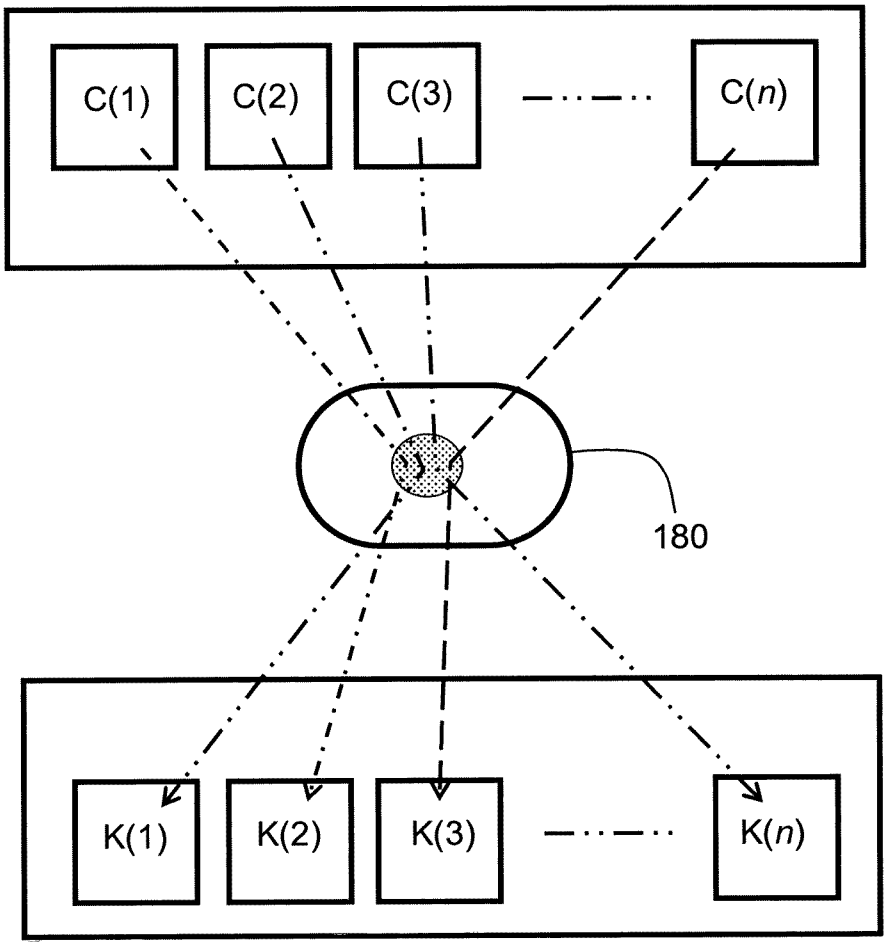


FIG. 3

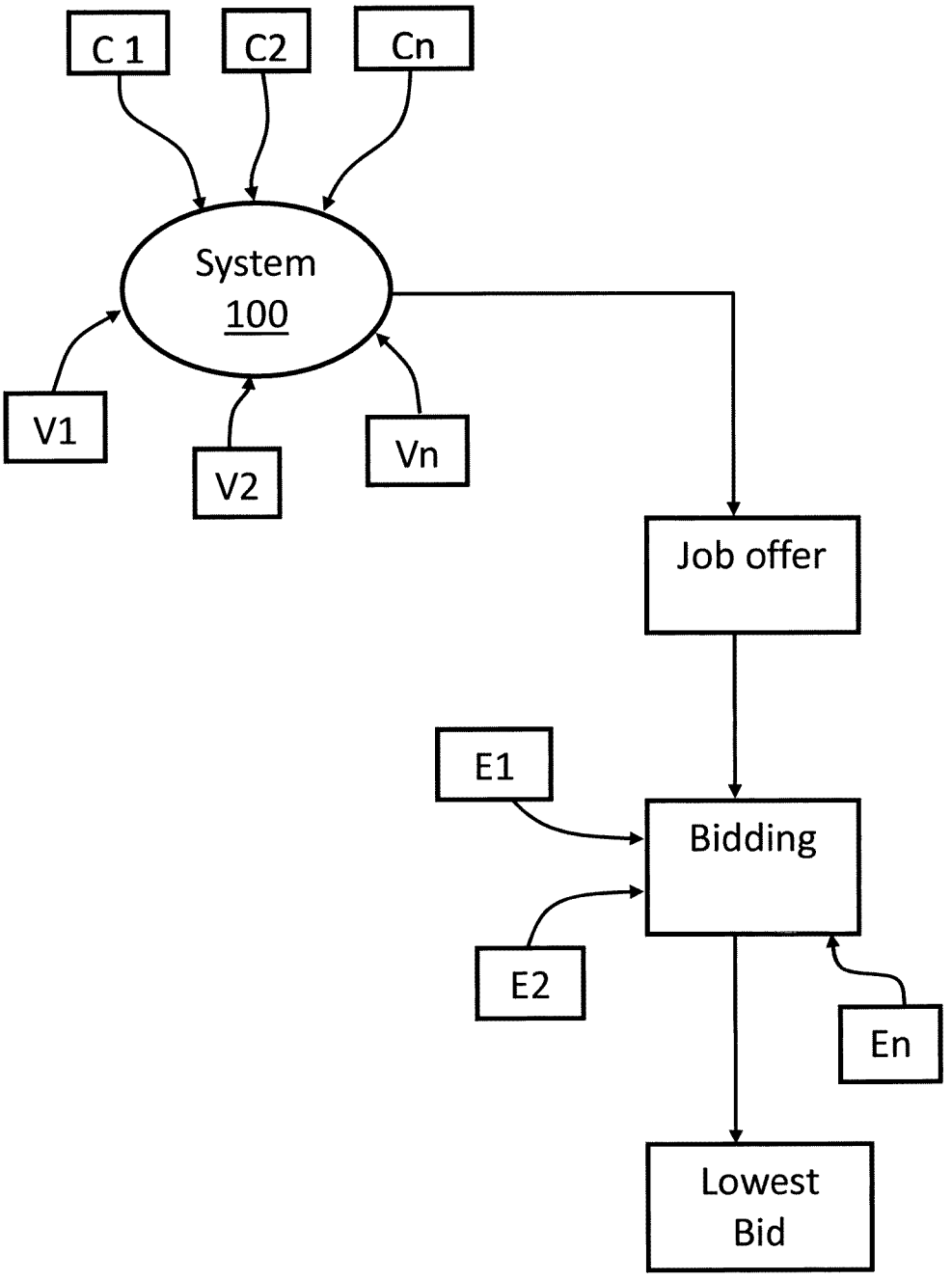

C 1
C2
Cn
System
100
V1
V2
Vn
Job offer
E1
Bidding
E2
En
Lowest
Bid

FIG. 4

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2013/003547

A. CLASSIFICATION OF SUBJECT MATTER
INV. G06Q10/06
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G06Q

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	DE 44 11 478 A1 (KRONE AG [DE]) 24 November 1994 (1994-11-24) abstract; claims 1-18; figure 1 column 1, line 18 - line 23 column 2, line 27 - line 37 column 3, line 1 - line 5 column 4, line 31 - line 32 column 4, line 59 - line 61 -----	1-19
X	EP 2 259 216 A1 (GITEN ENERGY MFG COMPANY LTD [TW]) 8 December 2010 (2010-12-08) claims 1-16; figure 4 paragraph [0030] - paragraph [0036] paragraph [0040] - paragraph [0042] ----- -/--	1-19

X Further documents are listed in the continuation of Box C.

X See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
17 March 2014	31/03/2014

Name and mailing address of the ISA/	Authorized officer
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1

Form PCT/ISA/210 (second sheet) (April 2005)

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2013/003547

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	WO 00/77696 A1 (TIETOKESKO OY [FI]; LEHTO JANNE [FI]; JAESKE TIMO [FI]; YLOENEN SEPPO) 21 December 2000 (2000-12-21) abstract; claims 1-11; figures 1-7D -----	1-19
A	GB 2 402 737 A (WTI EUROP LTD [GB]) 15 December 2004 (2004-12-15) claims 1-11; figure 1 -----	1-19

1

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/EP2013/003547

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